DICOM Change Proposal

STATUS	Letter Ballot
Date of Last Update	2026-03-22
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Change Number	CP-2595
Log Summary:	Extend KOS to support more metadata for manifest use case

Name of Standard

PS3.3, PS3.6, PS3.15, PS3.16

Rationale for Change:

The KOS has been extensively utilized in the context of cross-enterprise document sharing, including for images, particularly in IHE XDS-I.

There is a desire to allow selection of a subset of images described in a KOS rather than retrieving an entire set of images (such as a Study), based on metadata included in the manifest, or performing a query by some other mechanism.

Though TID 2010 required for the KOS IOD and Storage SOP Class is defined as non-extensible, it is thought to be preferrable to allow its extension in a non-breaking manner, rather than to sanction the use of extension of DICOM data set attributes in a manner that would not be in keeping with the design of the KOS IOD.

Inclusion of the Image Library sub-template, which is already defined for the purpose of describing referenced images in more detail, allows for sufficient but constrained extensibility in keeping with the spirit of the KOS design, albeit requiring (a) parsing of the Content Tree and (b) replication of the image references (in order to preserve compatibility with the existing flat list of image references in the TID 2010).

It is proposed to add TID 1600 “Image Library” as an additional row to TID 2010, and requiring its presence if a specified (new) Document Title code is used.

The Image Library sub-templates are extended to include some of the metadata proposed for the extended manifest based on experience of what is useful for selecting Series from Studies for viewing, and without harming their current use in other templates that already make use of them.

The Image Library sub-templates are extensible, and their contents are mostly optional, so for specific-use cases, additional constraints may be specified in an externally defined profile (such as in IHE MADO), which might require the presence of specific Content Items, and/or the use of specific value sets.

In addition, this CP introduces a Display URI Attribute in the General Study Module to provide a mechanism for viewer invocation. While existing Retrieve Location UID and Retrieve URL Attributes support object retrieval, they do not define semantics for launching a remote image display application. In cross-enterprise and document sharing environments, it is common to provide an endpoint capable of invoking a viewer with appropriate context.

Change Wording:

Modify PS3.3 Section C.7.2.1 General Study Module, as follows:

C.7.2.1 General Study Module

…

Table C.7-3. General Study Module Attributes

Attribute Name	Tag	Type	Attribute Description
…	…	…	…
Display URI	**(gggg,eeee)**	**3**	**URI specifying the access path to a remote image display service for the Study.** **Includes fully specified scheme, authority, path, and query components in accordance with [RFC3986].**

For reference, relevant SR definitions from PS3.3:

3.21 Structured Report Definitions

Content Item	A node in the Content Tree of a DICOM SR document, consisting of either a container with a coded Concept Name, or a name-value pair with a coded Concept Name and a Concept Value.
Content Tree	The tree of Content Items of a DICOM SR document.

…

C.17.3.1 SR Document Content Tree

The Module consists of a single root Content Item that is the root of the SR Document Content Tree. The root Content Item is of type CONTAINER, and its Content Sequence conveys either directly or indirectly through further nested Content Sequences, all of the other Content Items in the document. This root Content Item shall have a heading in the Concept Name Code Sequence (0040,A043) that conveys the title of the SR Document, i.e., the Document Title.

…

C.17.2.3 Current Requested Procedure Evidence Sequence and Pertinent Other Evidence Sequence

The intent of the Current Requested Procedure Evidence Sequence (0040,A375) is to reference all evidence created in order to satisfy the current Requested Procedure(s) for this SR Document. This shall include, but is not limited to, all current evidence referenced in the Content Tree.

For a completed SR Document satisfying (i.e., being the final report for) the current Requested Procedure(s), this Sequence shall list the full set of Composite SOP Instances created for the current Requested Procedure(s). For other SOP Instances that include the SR Document General Module, this Sequence shall contain at minimum the set of Composite SOP Instances from the current Requested Procedure(s) that are referenced in the Content Tree.

The Pertinent Other Evidence Sequence (0040,A385) is used to reference all other evidence considered pertinent for this SR Document that is not listed in the Current Requested Procedure Evidence Sequence (0040,A375).

This requires that the same SOP Instance shall not be referenced in both of these Sequences.

For the purposes of inclusion in the Current Requested Procedure Evidence Sequence (0040,A375) and the Pertinent Other Evidence Sequence (0040,A385), the set of Composite SOP Instances is defined to include not only the images and waveforms referenced in the Content Tree, but also all presentation states, Real World Value maps and other accompanying Composite Instances that are referenced from the Content Items.

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Add a new subsection to PS3.3 C.17.6.2 Key Object Document Module as indicated

(changes to existing text are bold and underlined for additions and bold and struckthrough for removals):

C.17.6.2 Key Object Document Module

...

Table C.17.6-2. Key Object Document Module Attributes

Attribute Name	Tag	Type	Attribute Description
…	…	…	…
Current Requested Procedure Evidence Sequence	(0040,A375)	1	List of all Composite SOP Instances referenced in the Content Sequence (0040,A730). ~~**The current evidence is considered to be only the set of Instances referenced within the Key Object Selection.**~~ **See Section C.17.6.2.n.** One or more Items shall be included in this Sequence.
>Include Table C.17-3 "Hierarchical SOP Instance Reference Macro Attributes"			
Identical Documents Sequence	(0040,A525)	1C	Duplicates of this document, stored with different SOP Instance UIDs. One or more Items shall be included in this Sequence. Required if this Key Object Selection document references Instances in more than one Study. See Section C.17.2.2 and Section C.17.6.2.1 for further explanation and conditions.
>Include Table C.17-3 "Hierarchical SOP Instance Reference Macro Attributes"			

C.17.6.2.1 Identical Documents

If the Current Requested Procedure Evidence Sequence (0040,A375) references SOP Instances both in the current Study and in one or more other Studies, this document shall be duplicated into each of those other Studies, and the duplicates shall be referenced in the Identical Documents Sequence (0040,A525).

Note

Thus a Key Object Selection Document that references images in the current Study as well as in a prior or comparison Study, would be duplicated into the other Study. This allows an application displaying that other Study to easily access notes relevant to that Study's SOP Instances.

C.17.6.2.n Current Requested Procedure Evidence Sequence

If the Document Title in the Concept Name Code Sequence (0040,A043) is (ddd001, DCM, "Manifest with Description"), then the Current Requested Procedure Evidence Sequence (0040,A375) shall reference all the SOP Instances that are being shared (i.e., the complete manifest), rather than only the set of Composite SOP Instances that are referenced in the Content Tree (see C.17.3.1).

If the Document Title in the Concept Name Code Sequence (0040,A043) is not (ddd001, DCM, "Manifest with Description"), then the current evidence is considered to be only the set of Instances referenced within the Key Object Selection.

Modify PS3.3 as indicated to expand the relationship constraints to support the Image Library

(changes to existing text are bold and underlined for additions and bold and struckthrough for removals):

A.35.4 Key Object Selection Document IOD

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A.35.4.3.1 Key Object Selection Document IOD Content Constraints

A.35.4.3.1.1 Value Type

Value Type (0040,A040) in Content Sequence (0040,A730) of the SR Document Content Module is constrained to the following Enumerated Values (see Table C.17.3-7 for Value Type definitions):
Enumerated Values:
TEXT
CODE
UIDREF
PNAME
COMPOSITE
IMAGE
WAVEFORM
CONTAINER
<u>DATE</u>
<u>TIME</u>
<u>DATETIME</u>
<u>NUM</u>

A.35.4.3.1.2 Relationship Constraints

Relationships between Content Items in the content of this IOD shall be conveyed in the by-value mode. See Table C.17.3-8 for Relationship Type definitions.

Note
Relationships by-reference are forbidden. Therefore, Referenced Content Item Identifier (0040,DB73) is not present in any of the Content Items within the SR Document Content Module.

Table A.35.4-2 specifies the relationship constraints of this IOD.

Table A.35.4-2. Relationship Content Constraints for Key Object Selection Document IOD

Source Value Type	Relationship Type (Enumerated Values)	Target Value Type
CONTAINER	CONTAINS	TEXT, IMAGE, WAVEFORM, COMPOSITE**<u>,</u>** **<u>CONTAINER</u>**
CONTAINER	HAS OBS CONTEXT	TEXT, CODE, UIDREF, PNAME, CONTAINER
CONTAINER	HAS CONCEPT MOD	CODE
<u>CONTAINER</u>	**<u>HAS ACQ CONTEXT</u>**	**<u>CODE, DATE, TIME, DATETIME, UIDREF, NUM, TEXT</u>**
<u>IMAGE</u>	**<u>HAS ACQ CONTEXT</u>**	**<u>CODE, DATE, TIME, DATETIME, UIDREF, NUM, TEXT</u>**
<u>COMPOSITE</u>	**<u>HAS ACQ CONTEXT</u>**	**<u>CODE, DATE, TIME, DATETIME, UIDREF, NUM, TEXT</u>**
<u>WAVEFORM</u>	**<u>HAS ACQ CONTEXT</u>**	**<u>CODE, DATE, TIME, DATETIME, UIDREF, NUM, TEXT</u>**

Note
The SOP Classes to which an IMAGE, WAVEFORM or COMPOSITE Value Type may refer are documented in the Conformance Statement for an application (see PS3.2 and PS3.4).

A.35.4.3.1.3 Template Constraints

The document shall be constructed from TID 2010 “Key Object Selection” invoked at the root node.

Modify PS3.6 Table 6-1. Registry of DICOM Data Elements, as follows:

Table 6-1. Registry of DICOM Data Elements

Tag	Name	Keyword	VR	VM	
…					
(gggg,eeee)	**Display URI**	**DisplayURI**	**UR**	**1**	
…					

Modify PS3.15 Table E.1-1. Application Level Confidentiality Profile Attributes, as follows:

Table E.1-1. Application Level Confidentiality Profile Attributes

Attribute Name	Tag	Retd. (from PS3.6)	In Std. Comp. IOD (from PS3.3)	Basic Prof.	Rtn. Safe Priv. Opt.	Rtn. UIDs Opt.	Rtn. Dev. Id. Opt.	Rtn. Inst. Id. Opt.	Rtn. Pat. Chars. Opt.	Rtn. Long. Full Dates Opt.	Rtn. Long. Modif. Dates Opt.	Clean Desc. Opt.	Clean Struct. Cont. Opt.	Clean Graph. Opt.
…														
Display URI	**(gggg,eeee)**	**N**	**Y**	**X**										
…														

Modify PS3.15 E.3.4-1. Application Level Confidentiality Profile Clean Structured Content Option Content Item Concept Name Codes, as follows:

Table E.3.4-1. Application Level Confidentiality Profile Clean Structured Content Option Content Item Concept Name Codes

Code Meaning	Code Value	Coding Scheme Designator	Value Type	Retd. (from PS3.16)	In Std. Tpl. (from PS3.16)	Basic Prof.	Rtn. UIDs Opt.	Rtn. Dev. Id. Opt.	Rtn. Inst. Id. Opt.	Rtn. Pat. Chars. Opt.	Rtn. Long. Full Dates Opt.	Rtn. Long. Modif. Dates Opt.	Clean Desc. Opt.
…													
Series Date	**ddd003**	**DCM**	**DATE**	**N**	**Y**	**X**					**K**	**C**	
Series Description	**ddd002**	**DCM**	**TEXT**	**N**	**Y**	**X**							**C**
Series Time	**ddd004**	**DCM**	**TIME**	**N**	**Y**	**X**					**K**	**C**	
Series Instance UID	112002	DCM	UIDREF	N	Y	D	K						
…													

Modify PS3.16 TID 2010 Table as indicated, to include Image Library

(changes to existing text are bold and underlined for additions and bold and struckthrough for removals):

TID 2010 Key Object Selection

The Key Object Selection Template is intended for flagging one or more significant images, waveforms, or other composite SOP Instances. Key Object Selection contains:

- coded document title stating the reason for significance of the referenced objects in the Key Object Selection,

- optional free form text comment in an explicitly identified language, ~~and~~
- optional identification of the observer (device or person) that created the Key Object Selection~~.~~**, and**
- **optional Image Library to further describe the related instances**.

Note

1. For instance, when this Template is used to identify images rejected for quality reasons, the device or person performing the quality assessment is identified in observation context items (invoked through TID 1002 “Observer Context”). The reason for rejection can be included both as a code used as a concept modifier for the document title, and as text description.
2. The order of object references may be significant, e.g., when the title concept is "For Conference".
3. Instances referenced in a Key Object Selection Document may be securely referenced by Digital Signature or MAC mechanisms within the SR Document General Module (see PS3.3).

The Template can only be instantiated at the root node and cannot be included in other Templates. The Template is not extensible; that is, no other Content Items may be added to this Template, or the Templates that are included, recursively**, except for TID 1600 “Image Library”, which may be extended**.

Type: **Non-Extensible**
Order: **Non-Significant**
Root: **Yes**

Table TID 2010. Key Object Selection

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1			CONTAINER	DCID 7010 “Key Object Selection Document Title”	1	M		Root node
2	>	HAS CONCEPT MOD	CODE	EV (113011, DCM, "Document Title Modifier")	1-n	U		
3	>	HAS CONCEPT MOD	CODE	EV (113011, DCM, "Document Title Modifier")	1	UC	IF Row 1 Concept Name = (113001, DCM, "Rejected for Quality Reasons") or (113010, DCM, "Quality Issue")	DCID 7011 “Rejected for Quality Reason”
4	>	HAS CONCEPT MOD	CODE	EV (113011, DCM, "Document Title Modifier")	1	MC	IF Row 1 Concept Name = (113013, DCM, "Best In Set")	DCID 7012 “Best in Set”
4b	**>**	**HAS CONCEPT MOD**	**CODE**	**EV (121023, DCM “Procedure Code”)**	**1-n**	**U**		
5	>	HAS CONCEPT MOD	INCLUDE	DTID 1204 “Language of Content Item and Descendants”	1	U		
6	>	HAS OBS CONTEXT	INCLUDE	DTID 1002 “Observer Context”	1-n	U		
7	>	CONTAINS	TEXT	EV (113012, DCM, "Key Object Description")	1	U		
8	>	CONTAINS	IMAGE		1-n	MC	At least one of Rows 8, 9 and 10 shall be present	
9	>	CONTAINS	WAVEFORM		1-n	MC	At least one of Rows 8, 9 and 10 shall be present	
10	>	CONTAINS	COMPOSITE		1-n	MC	At least one of Rows 8, 9 and 10 shall be present	
11	**>**	**CONTAINS**	**INCLUDE**	**DTID 1600 “Image Library”**	**1**	**MC**	**IFF Row 1 Concept Name = (ddd001, DCM, "Manifest with Description")**	

Content Item Descriptions

Row 4b	**The Procedure Code represents the procedure (scheduled or performed) associated with the referenced SOP Instances.**
Rows 8, 9, 10	Purpose of reference shall not be present. **All SOP Instances referenced in the Current Requested Procedure Evidence Sequence (0040,A375) shall also be referenced in Rows 8, 9 and 10.** **When the Row 1 Concept Name is (ddd001, DCM, "Manifest with Description") then all referenced instances shall be in the same Study as the KOS.**
Row 11	**Though TID 2010 itself is non-extensible, the included TID 1600 is extensible.** **The Image Library may contain an entry for any instance referenced in Rows 8, 9 and 10, and not include entries for instances not referenced in those Rows. For example, a representative image from each Series might be described.**

Modify PS3.16 TID 1600 and sub-templates as indicated, to add metadata needed for extended manifest

(changes to existing text are bold and underlined for additions and bold and struckthrough for removals):

TID 1600 Image Library

The Image Library contains references to images and selected attributes describing them that facilitate analysis without having to retrieve the entire set of referenced images.

Note

An Image Library Entry may reference IMAGE, COMPOSITE, or WAVEFORM SOP Instances.
The name of this Template reflects its original scope and is retained for backward compatibility.

Type: **Extensible**
Order: **Non-Significant**
Root: **No**

Table TID 1600. Image Library

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1			CONTAINER	EV (111028, DCM, "Image Library")	1	M		
1b	**>**	**HAS ACQ CONTEXT**	**CODE**	**EV (121139, DCM, "Modality")**	**1-n**	**U**		**DCID 33 "Modality"**
1c	**>**	**HAS ACQ CONTEXT**	**UIDREF**	**EV (110181, DCM, "SOP Class UID")**	**1-n**	**U**		
1d	**>**	**HAS ACQ CONTEXT**	**CODE**	**EV (123014, DCM, "Target Region")**	**1-n**	**UC**	**XOR with Row 1e**	**DCID 4031 "Common Anatomic Region"**
1e	**>**	**HAS ACQ CONTEXT**	**TEXT**	**EV (123014, DCM, "Target Region")**	**1-n**	**UC**	**XOR with Row 1d**	
1f	**>**	**HAS ACQ CONTEXT**	**NUM**	**EV (ddd009, DCM, "Number of Study Related Series"**	**1**	**U**		**UNITS = EV ({series}, UCUM, "series")**
2	>	CONTAINS	CONTAINER	EV (126200, DCM, "Image Library Group")	1-n	U		
3	>>	HAS ACQ CONTEXT	INCLUDE	DTID 1602 "Image Library Entry Descriptors"	1	U		
4	>>	CONTAINS	INCLUDE	DTID 1601 "Image Library Entry"	1-n	U		

Content Item Descriptions

Row 1b	**The Modality may be specified here, at the Image Library level, to avoid specifying redundant information in Image Library Entries. If specified, all of the Modalities within the Study that are referenced are specified, similar to the PS3.4 Modalities in Study (0008,0061) additional Query/Retrieve Attribute.**

Row 1c	**SOP Class UID may be specified here, at the Image Library level, to avoid specifying redundant information in Image Library Entries. If specified, all of the SOP Class UIDs within the Study that are referenced are specified, similar to the PS3.4 SOP Classes in Study (0008,0062) additional Query/Retrieve Attribute.**
Row 1d, 1e	**Target Region may be specified here, at the Image Library level, to avoid specifying redundant information in Image Library Entries. If specified, all of the Target Regions within the Study that are referenced are specified.** **The Target Region may be a text value (e.g., the Value of Body Part Examined (0018,0015), and/or a code value.**
Row 1f	**This shall be the number of Series of Instances listed in Rows 8, 9 or 10 of TID 2010.**
Row 2	**Any grouping may be defined, e.g., all SOP Instances within a single Series may be included in a single Image Library Group.**
Row 3	These Image Library Entry Descriptors apply to all Image Library Entries in this Image Library Group.

TID 1601 Image Library Entry

Each ~~instance~~**invocation** of the Image Library Entry Template contains the ~~Image~~ SOP Class and Instance UIDs, and selected attributes ~~for an image~~ that facilitate analysis without having to retrieve the entire set of referenced ~~images~~ **instances**.

Type: **Extensible**
Order: **Non-Significant**
Root: **No**

Table TID 1601. Image Library Entry

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1			IMAGE		1	M**C**	**XOR Rows 3, 5**	
2	>	HAS ACQ CONTEXT	INCLUDE	DTID 1602 “Image Library Entry Descriptors”	1	U		
3			**COMPOSITE**		**1**	**MC**	**XOR Rows 1, 5**	
4	**>**	**HAS ACQ CONTEXT**	**INCLUDE**	**DTID 1602 “Image Library Entry Descriptors”**	**1**	**U**		
5			**WAVEFORM**		**1**	**MC**	**XOR Rows 1, 3**	
6	**>**	**HAS ACQ CONTEXT**	**INCLUDE**	**DTID 1602 “Image Library Entry Descriptors”**	**1**	**U**		

Content Item Descriptions

Row**s** 2**, 4, 6**	These Image Library Entry Descriptors apply to the IMAGE**, COMPOSITE or WAVEFORM** in Row**s** 1**, 3 and 5,** and override descriptors in Row 3 of Section TID 1600 in case of conflict.

TID 1602 Image Library Entry Descriptors

This Template contains selected attributes for an ~~image~~**instance** or group of ~~images~~**instances**. The descriptive information may be copied from ~~images~~**instances** or derived.

Type: **Extensible**
Order: **Non-Significant**
Root: **No**

Table TID 1602. Image Library Entry Descriptors

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1		HAS ACQ CONTEXT	CODE	EV (121139, DCM, "Modality")	1	U		~~DCID 29 “Acquisition Modality”~~ **DCID 33 “Modality”**
2		HAS ACQ CONTEXT	CODE	EV (123014, DCM, "Target Region")	1	U**C**	**XOR with Row 2b**	DCID 4031 “Common Anatomic Region”
2b		**HAS ACQ CONTEXT**	**TEXT**	**EV (123014, DCM, "Target Region")**	**1**	**UC**	**XOR with Row 2**	

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
3		HAS ACQ CONTEXT	CODE	EV (111027, DCM, "Image Laterality")	1	U		DCID 244 “Laterality”
4		HAS ACQ CONTEXT	DATE	EV (111060, DCM, "Study Date")	1	U		
5		HAS ACQ CONTEXT	TIME	EV (111061, DCM, "Study Time")	1	U		
5a		**HAS ACQ CONTEXT**	**UIDREF**	**EV (112002, DCM, “Series Instance UID”)**	**1**	**U**		
5b		**HAS ACQ CONTEXT**	**TEXT**	**EV (113607, DCM, “Series Number”)**	**1**	**U**		
5c		**HAS ACQ CONTEXT**	**TEXT**	**EV (ddd002, DCM, “Series Description”)**	**1**	**UC**	**XOR with Row 5d**	
5d		**HAS ACQ CONTEXT**	**CODE**	**EV (ddd002, DCM, “Series Description”)**	**1**	**UC**	**XOR with Row 5c**	
5e		**HAS ACQ CONTEXT**	**DATE**	**EV (ddd003, DCM, "Series Date")**	**1**	**U**		
5f		**HAS ACQ CONTEXT**	**TIME**	**EV (ddd004, DCM, "Series Time")**	**1**	**U**		
5g		**HAS ACQ CONTEXT**	**NUM**	**EV (ddd007, DCM, “Number of Series Related Instances”)**	**1**	**U**		**UNITS = EV ({instances}, UCUM, "instances")**
6		HAS ACQ CONTEXT	DATE	EV (111018, DCM, "Content Date")	1	U		
7		HAS ACQ CONTEXT	TIME	EV (111019, DCM, "Content Time")	1	U		
8		HAS ACQ CONTEXT	DATE	EV (126201, DCM, "Acquisition Date")	1	U		
9		HAS ACQ CONTEXT	TIME	EV (126202, DCM, "Acquisition Time")	1	U		
10		HAS ACQ CONTEXT	UIDREF	EV (112227, DCM, "Frame of Reference UID")	1	U		
11		HAS ACQ CONTEXT	NUM	EV (110910, DCM, "Pixel Data Rows")	1	U		UNITS = EV ({pixels}, UCUM, "pixels")
12		HAS ACQ CONTEXT	NUM	EV (110911, DCM, "Pixel Data Columns")	1	U		UNITS = EV ({pixels}, UCUM, "pixels")
12a		**HAS ACQ CONTEXT**	**TEXT**	**EV (113609, DCM, “Instance Number”)**	**1**	**U**		
12b		**HAS ACQ CONTEXT**	**NUM**	**EV (121140, DCM, “Number of Frames”)**	**1**	**U**		**UNITS = EV ({frames}, UCUM, "frames")**
13		HAS ACQ CONTEXT	INCLUDE	DTID 1603 “Image Library Entry Descriptors for Projection Radiography”	1	MC	IFF Row 1 is present with a value of "CR", "DX", "IO", "MG, "PX", "RF", "RG" or "XA"	
14		HAS ACQ CONTEXT	INCLUDE	DTID 1604 “Image Library Entry Descriptors for Cross-Sectional Modalities”	1	MC	IFF Row 1 is present with a value of "CT", "MR" or "PT"	
15		HAS ACQ CONTEXT	INCLUDE	DTID 1605 “Image Library Entry Descriptors for CT”	1	MC	IFF Row 1 is present with a value of "CT"	
16		HAS ACQ CONTEXT	INCLUDE	DTID 1606 “Image Library Entry Descriptors for MR”	1	MC	IFF Row 1 is present with a value of "MR"	

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
17		HAS ACQ CONTEXT	INCLUDE	DTID 1607 “Image Library Entry Descriptors for PET”	1	MC	IFF Row 1 is present with a value of "PT"	
18		**HAS ACQ CONTEXT**	**INCLUDE**	**DTID 16xx “Image Library Entry Descriptors for Key Object Selection”**	**1**	**MC**	**IFF Row 1 is present with a value of "KO"**	

Content Item Descriptions

Row 2**, 2b**	**The Target Region may be a text value (e.g., the Value of Body Part Examined (0018,0015)), or a code, (e.g., t**The value of Anatomic Region Sequence (0008,2218)), in the Image IOD, or a code derived from Body Part Examined (0018,0015) using the mapping described in Annex L.
Row 5c, 5d	**The Series Description may be a text value (e.g., the Value of Series Description (0008,103E)), and/or a code, (e.g., the Value of Series Description Code Sequence (0008,103F)).**
Row 5g	**This shall be the number of Series Related Instances listed in Rows 8, 9 or 10 of TID 2010.**

TID 1603 Image Library Entry Descriptors for Projection Radiography

This Template contains selected attributes for a projection radiography image or group of such images. The descriptive information may be copied from images or derived.

Type: **Extensible**
Order: **Non-Significant**
Root: **No**

Table TID 1603. Image Library Entry Descriptors for Projection Radiography

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1		HAS ACQ CONTEXT	CODE	EV (111031, DCM, "Image View")	1	U		
2	>	HAS CONCEPT MOD	CODE	EV (111032, DCM, "Image View Modifier")	1-n	U		
3		HAS ACQ CONTEXT	TEXT	EV (111044, DCM, "Patient Orientation Row")	1	U		
4		HAS ACQ CONTEXT	TEXT	EV (111043, DCM, "Patient Orientation Column")	1	U		
5		HAS ACQ CONTEXT	NUM	EV (111026, DCM, "Horizontal Pixel Spacing")	1	U		UNITS = EV (mm, UCUM, "millimeter")
6		HAS ACQ CONTEXT	NUM	EV (111066, DCM, "Vertical Pixel Spacing")	1	U		UNITS = EV (mm, UCUM, "millimeter")
7		HAS ACQ CONTEXT	NUM	EV (112011, DCM, "Positioner Primary Angle")	1	U		UNITS = EV (deg, UCUM, "deg")
8		HAS ACQ CONTEXT	NUM	EV (112012, DCM, "Positioner Secondary Angle")	1	U		UNITS = EV (deg, UCUM, "deg")

Content Item Descriptions

Rows 3, 4	First (row) and second (column) components of Patient Orientation (0020,0020) in the Image IOD. See Section C.7.6.1.1.1 “Patient Orientation” in PS3.3.
Row 5	The second component of Imager Pixel Spacing (0018,1164) in the Image IOD. See Section C.8.11.4 “DX Detector Module” in PS3.3.
Row 6	The first component of Imager Pixel Spacing (0018,1164) in the Image IOD. See Section C.8.11.4 “DX Detector Module” in PS3.3.

TID 1604 Image Library Entry Descriptors for Cross-Sectional Modalities

This Template contains selected attributes for a cross-sectional image or group of such images. The descriptive information may be copied from images or derived.

Type: **Extensible**
Order: **Non-Significant**

Root: No

Table TID 1604. Image Library Entry Descriptors for Cross-Sectional Modalities

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1		HAS ACQ CONTEXT	NUM	EV (111026, DCM, "Horizontal Pixel Spacing")	1	U		UNITS = EV (mm, UCUM, "millimeter")
2		HAS ACQ CONTEXT	NUM	EV (111066, DCM, "Vertical Pixel Spacing")	1	U		UNITS = EV (mm, UCUM, "millimeter")
3		HAS ACQ CONTEXT	NUM	EV (112226, DCM, "Spacing between slices")	1	U		UNITS = EV (mm, UCUM, "millimeter")
4		HAS ACQ CONTEXT	NUM	EV (112225, DCM, "Slice Thickness")	1	U		UNITS = EV (mm, UCUM, "millimeter")
5		HAS ACQ CONTEXT	NUM	EV (110901, DCM, "Image Position (Patient) X")	1	U		UNITS = EV (mm, UCUM, "millimeter")
6		HAS ACQ CONTEXT	NUM	EV (110902, DCM, "Image Position (Patient) Y")	1	U		UNITS = EV (mm, UCUM, "millimeter")
7		HAS ACQ CONTEXT	NUM	EV (110903, DCM, "Image Position (Patient) Z")	1	U		UNITS = EV (mm, UCUM, "millimeter")
8		HAS ACQ CONTEXT	NUM	EV (110904, DCM, "Image Orientation (Patient) Row X")	1	U		UNITS = EV ({-1:1}, UCUM, "{-1:1}")
9		HAS ACQ CONTEXT	NUM	EV (110905, DCM, "Image Orientation (Patient) Row Y")	1	U		UNITS = EV ({-1:1}, UCUM, "{-1:1}")
10		HAS ACQ CONTEXT	NUM	EV (110906, DCM, "Image Orientation (Patient) Row Z")	1	U		UNITS = EV ({-1:1}, UCUM, "{-1:1}")
11		HAS ACQ CONTEXT	NUM	EV (110907, DCM, "Image Orientation (Patient) Column X")	1	U		UNITS = EV ({-1:1}, UCUM, "{-1:1}")
12		HAS ACQ CONTEXT	NUM	EV (110908, DCM, "Image Orientation (Patient) Column Y")	1	U		UNITS = EV ({-1:1}, UCUM, "{-1:1}")
13		HAS ACQ CONTEXT	NUM	EV (110909, DCM, "Image Orientation (Patient) Column Z")	1	U		UNITS = EV ({-1:1}, UCUM, "{-1:1}")

Content Item Descriptions

Row 1	The second component of Pixel Spacing (0028,0030) in the Image IOD. See Section 10.7.1.1 “Pixel Spacing” in PS3.3 and Section C.7.6.2 “Image Plane Module” in PS3.3.
Row 2	The first component of Pixel Spacing (0028,0030) in the Image IOD. See Section 10.7.1.1 “Pixel Spacing” in PS3.3 and Section C.7.6.2 “Image Plane Module” in PS3.3.

TID 1605 Image Library Entry Descriptors for CT

This Template contains selected attributes for a CT image or group of such images. The descriptive information may be copied from images or derived.

Type: Extensible
Order: Non-Significant
Root: No

Table TID 1605. Image Library Entry Descriptors for CT

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1		HAS ACQ CONTEXT	CODE	EV (113820, DCM, "CT Acquisition Type")	1	U		DCID 10013 “CT Acquisition Type”
2		HAS ACQ CONTEXT	CODE	EV (113961, DCM, "Reconstruction Algorithm")	1	U		DCID 10033 “CT Reconstruction Algorithm”

Content Item Descriptions

Row 1	A code derived from the Value of Acquisition Type (0018,9302) in the Image IOD. See Section C.8.15.3.2 “CT Acquisition Type Macro” in PS3.3.

Row 2	A code derived from the Value of Reconstruction Algorithm (0018,9315) in the Image IOD. See Section C.8.15.3.7 “CT Reconstruction Macro” in PS3.3.

TID 1606 Image Library Entry Descriptors for MR

This Template contains selected attributes for a MR image or group of such images. The descriptive information may be copied from images or derived. Specialized coded Content Items allow more precise description of imaging sequences used for interpretation of multiparametric prostate MRI.

Type: **Extensible**
Order: **Non-Significant**
Root: **No**

Table TID 1606. Image Library Entry Descriptors for MR

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1		HAS ACQ CONTEXT	TEXT	EV (128230, DCM, "Pulse Sequence Name")	1	U		
2		HAS ACQ CONTEXT	NUM	EV (130542, DCM, "Magnetic field strength")	1	U		UNITS = (T, UCUM, "Tesla")
3		HAS ACQ CONTEXT	NUM	EV (RID10813, RADLEX, "MR coil")	1-n	U		DCID 6349 “MR Coil Type”
4		HAS ACQ CONTEXT	NUM	EV (110852, DCM, "MR signal intensity")	1	U		BCID 6311 “MR Signal Intensity”
5		HAS ACQ CONTEXT	NUM	EV (130546, DCM, "Cross-sectional scan plane orientation")	1	U		BCID 6312 “Cross-sectional Scan Plane Orientation”
6		HAS ACQ CONTEXT	NUM	EV (113240, DCM, "Source image diffusion b-value")	1-n	U		UNITS = (s/mm2, UCUM, "s/mm2")
7		HAS ACQ CONTEXT	INCLUDE	DTID 1608 “Image Library Entry Descriptors for Prostate Multiparametric MR”	1	U		

Content Item Descriptions

Row 1	The value of Pulse Sequence Name (0018,9005) or Sequence Name (0018,0024) in the Image IOD. See Section C.8.13.4 “MR Pulse Sequence Module” in PS3.3.
Row 6	Multiple values may apply when entry descriptor corresponds to a parametric map such as Apparent Diffusion Coefficient (ADC) map, which utilizes multiple b-values from a Diffusion-Weighted acquisition.

TID 1607 Image Library Entry Descriptors for PET

This Template contains selected attributes for a PET image or group of such images. The descriptive information may be copied from images or derived.

Note
The content of this Template is similar to that in TID 15101 NM/PET Protocol Context, but is in the form of an SR Template rather than a Protocol Context Template, and the Content Items are not nested as modifiers. There is also some similarity to TID 3307 NM/PET Perfusion Measurement Group.

Type: **Extensible**
Order: **Non-Significant**
Root: **No**

Table TID 1607. Image Library Entry Descriptors for PET

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1		HAS ACQ CONTEXT	CODE	EV (89457008, SCT, "Radionuclide")	1	U		DCID 4020 “PET Radionuclide”
2		HAS ACQ CONTEXT	CODE	EV (349358000, SCT, "Radiopharmaceutical agent")	1	U		DCID 4021 “PET Radiopharmaceutical”
3		HAS ACQ CONTEXT	NUM	EV (304283002, SCT, "Half-life of radiopharmaceutical")	1	U		UNITS = EV (s, UCUM, "s")

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
3b		HAS ACQ CONTEXT	TEXT	EV (121022, DCM, "Accession Number")	1	U		
4		HAS ACQ CONTEXT	DATETIME	EV (123003, DCM, "Radiopharmaceutical Start DateTime")	1	U		
5		HAS ACQ CONTEXT	DATETIME	EV (123004, DCM, "Radiopharmaceutical Stop DateTime")	1	U		
6		HAS ACQ CONTEXT	NUM	EV (123005, DCM, "Radiopharmaceutical Volume")	1	U		UNITS = DT (cm3, UCUM, "cm3")
7		HAS ACQ CONTEXT	NUM	EV (123006, DCM, "Radionuclide Total Dose")	1	U		UNITS = DT (Bq, UCUM, "Bq")
8		HAS ACQ CONTEXT	NUM	EV (123007, DCM, "Radiopharmaceutical Specific Activity")	1	U		UNITS = DT (Bq/mol, UCUM, "Bq/mol")
9		HAS ACQ CONTEXT	CODE	EV (410675002, SCT, "Route of Administration")	1	U		BCID 11 “Administration Route”
10		HAS ACQ CONTEXT	NUM	EV (123009, DCM, "Radionuclide Syringe Counts")	1	U		UNITS = DT ({counts}/s, UCUM, "counts/s")
11		HAS ACQ CONTEXT	NUM	EV (123010, DCM, "Radionuclide Residual Syringe Counts")	1	U		UNITS = DT ({counts}/s, UCUM, "counts/s")
12		HAS ACQ CONTEXT	NUM	EV (126203, DCM, "PET Radionuclide Incubation Time")	1	U		UNITS = EV (min, UCUM, "min")
13		HAS ACQ CONTEXT	NUM	EV (14749-6, LN, "Glucose")	1	U		UNITS = EV (mmol/l, UCUM, "mmol/l")
14		HAS ACQ CONTEXT	DATE	EV (127857, DCM, "Glucose Measurement Date")	1	MC	IF Row 13 Glucose is present and does not contain Observation DateTime (0040,A032).	
15		HAS ACQ CONTEXT	TIME	EV (127858, DCM, "Glucose Measurement Time")	1	MC	IF Row 13 Glucose is present and does not contain Observation DateTime (0040,A032).	

Content Item Descriptions

Row 3	Half-life of radiopharmaceutical	The units for half life are chosen to be seconds, to match the units used for Radionuclide Half Life (0018,1075). See Section C.8.9.2 “PET Isotope Module” in PS3.3.
Row 14	Glucose Measurement Date	In an earlier edition of the Standard, an incorrect DCM code was used for this concept, which was already assigned as (109081, DCM, "Prospective gating").
Row 15	Glucose Measurement Time	In an earlier edition of the Standard, an incorrect DCM code was used for this concept, which was already assigned as (109082, DCM, "Retrospective gating").

TID 1608 Image Library Entry Descriptors for Prostate Multiparametric MR

This Template includes attributes for image library entries that define the type of the sequence, as needed for PI-RADS interpretation of multiparametric MRI, specify most important sequence-specific attributes, and provide a location for reporting prostate imaging and sequence-specific technical characteristics of the acquisition.

Note
A descriptor specific to prostate MRI and PI-RADS is provided to record Prostate DCE temporal resolution. This term follows the conventions used in the PI-RADS guidelines.

Type: **Extensible**
Order: **Non-Significant**
Root: **No**

Table TID 1608. Image Library Entry Descriptors for Prostate Multiparametric MR

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1		HAS ACQ CONTEXT	TEXT	EV (130544, DCM, "Endorectal coil type")	1	U		
2		HAS ACQ CONTEXT	CODE	EV (130545, DCM, "Inflatable endorectal coil fill substance")	1	U		DCID 6350 “Endorectal Coil Fill Substance”
3		HAS ACQ CONTEXT	NUM	EV (130547, DCM, "Dynamic contrast-enhanced temporal resolution")	1-n	U		UNITS = (s, UCUM, "second")

Modify PS3.16 Section A to add a new Image Library Entry template TID 16xx as indicated, to include new concepts for describing the properties of the flagging of instances with Key Object Selection SOP Class

TID 16xx Image Library Entry Descriptors for Key Object Selection

This Template includes attributes for image library entries that specify most important attributes for access to instances flagged by Key Object Selection.

Type: **Extensible**
Order: **Non-Significant**
Root: **No**

Table TID 16xx. Image Library Entry Descriptors for Key Object Selection

	NL	Rel with Parent	VT	Concept Name	VM	Req Type	Condition	Value Set Constraint
1		HAS ACQ CONTEXT	CODE	EV (121144, DCM, "Document Title")	1	M		
2		HAS ACQ CONTEXT	TEXT	EV (113012, DCM, "Key Object Description"	1	MC	Required when present in the referenced KOS Instance	

Modify PS3.16 CID 7010 as indicated, to include new concepts

(changes to existing text are bold and underlined for additions and bold and struckthrough for removals):

CID 7010 Key Object Selection Document Title

Resources: **HTML | FHIR JSON | FHIR XML | IHE SVS XML**
Keyword: **KeyObjectSelectionDocumentTitle**
FHIR Keyword: **dicom-cid-7010-KeyObjectSelectionDocumentTitle**
Type: **Extensible**
Version: **~~20190915~~yyyymmdd**
UID: **1.2.840.10008.6.1.490**

Table CID 7010. Key Object Selection Document Title

Coding Scheme Designator	Code Value	Code Meaning
DCM	113000	Of Interest
DCM	113001	Rejected for Quality Reasons
DCM	113002	For Referring Provider
…		

Coding Scheme Designator	Code Value	Code Meaning
DCM	113021	For Litigation
DCM	113030	Manifest
DCM	113031	Signed Manifest
DCM	**ddd001**	**Manifest with Description**
DCM	113032	Complete Study Content
…		

Modify PS3.16 Annex D as indicated, to include new concepts

(changes to existing text are bold and underlined for additions and bold and struckthrough for removals):

Table D-1. DICOM Controlled Terminology Definitions (Coding Scheme Designator "DCM" Coding Scheme Version "01")

Code Value	Code Meaning	Definition	Notes
113030	Manifest	A list of objects that have been exported out of one organizational domain into another domain. Typically, the first domain has no direct control over what the second domain will do with the objects.	
113031	Signed Manifest	A signed list of objects that have been exported out of one organizational domain into another domain, referenced securely with either Digital Signatures or MACs. Typically, the first domain has no direct control over what the second domain will do with the objects.	
ddd001	**Manifest with Description**	**A list of objects that have been selected by a source organization for sharing. The manifest includes representative descriptive metadata to support determining the relevance of the listed objects to the recipient.**	
111018	Content Date	The date the data creation started.	
111019	Content Time	The time the data creation started.	
111060	Study Date	Date on which the acquisition of the study information was started.	
111061	Study Time	Time at which the acquisition of the study information was started.	
126201	Acquisition Date	The date the acquisition of data started	
126202	Acquisition Time	The time the acquisition of data started	
ddd003	**Series Date**	**Date the Series started.**	
ddd004	**Series Time**	**Time the Series started.**	
ddd002	**Series Description**	**Description of the Series.**	
121139	Modality	Type of device, process or method used to acquire or derive data.	
121140	Number of Frames	Number of Frames in a multi-frame image.	
ddd007	**Number of Series Related Instances**	**Count of SOP Instances in a Series.**	
ddd009	**Number of Study Related Series**	**Count of Series in a Study.**	